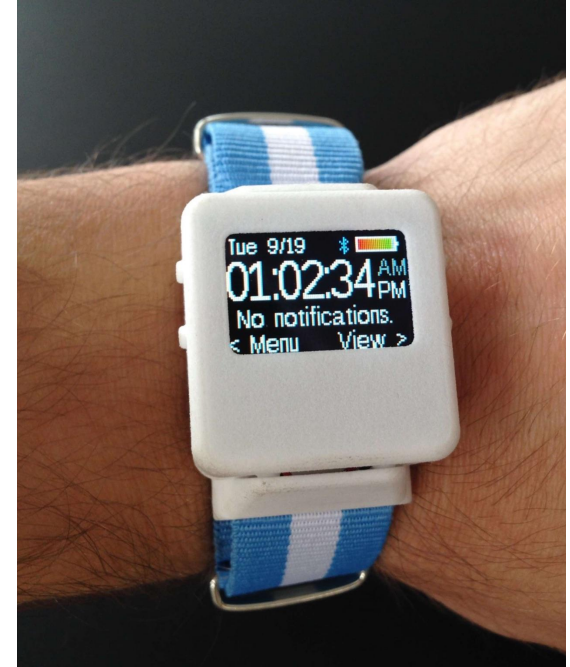
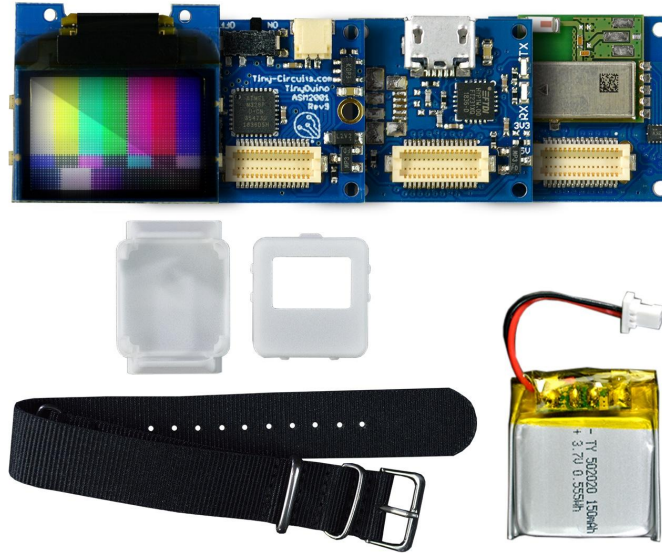


Tiny Watch



TinyDuino

- “The TinyCircuits TinyDuino is an Arduino compatible board in an ultra compact package. Imagine the possibilities of having the full power of an Arduino Uno in a size less than a quarter!”

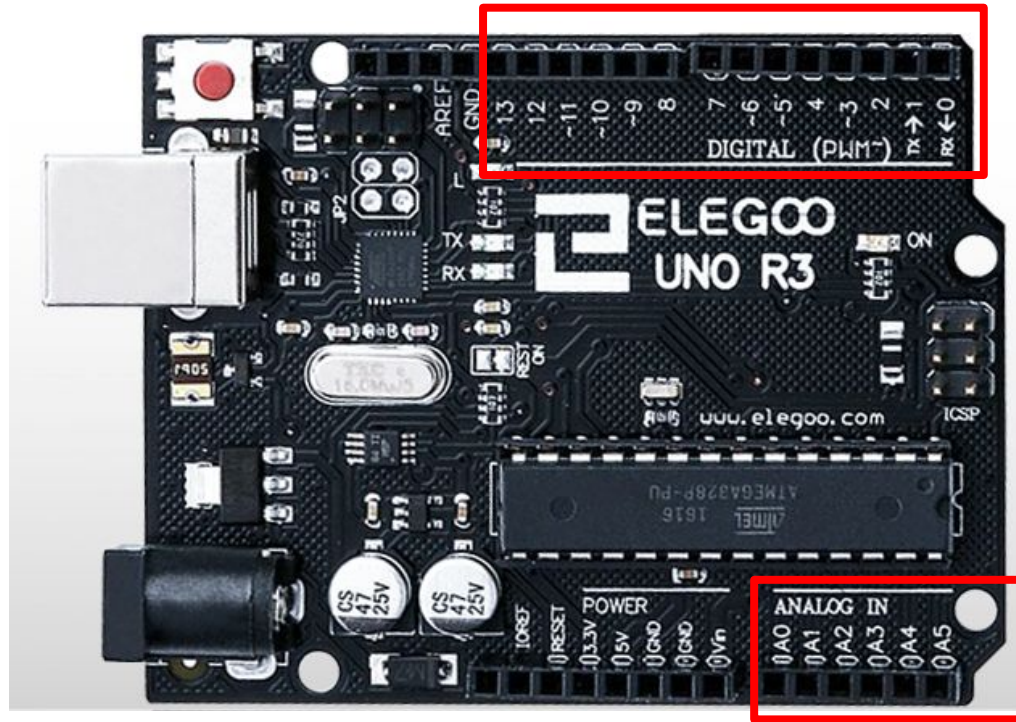


TinyDuino

Text below trimmed from TinyCircuits. URL: <https://tinycircuits.com/pages/tinyduino-overview>

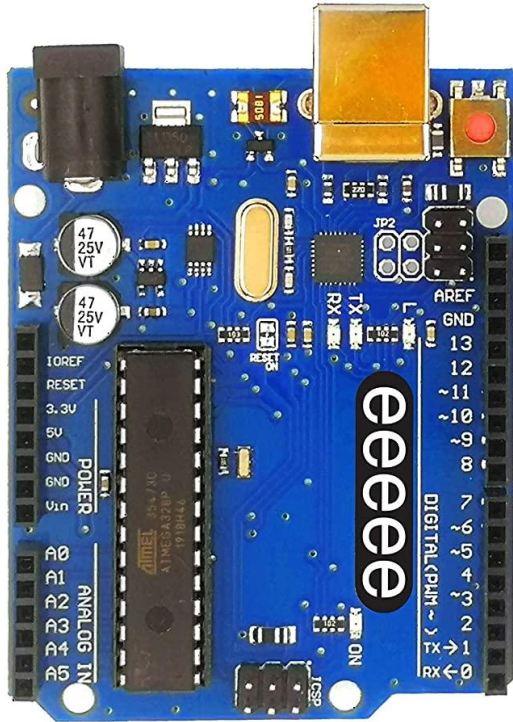
- The TinyDuino Platform is a miniature open-source electronics platform based on the easy-to-use hardware and software Arduino platform
- The platform is comprised of a TinyDuino processor board and multiple TinyShields which add special functions, like sensors, communications and display options
- By stacking the boards (with the tan 32-pin connectors), you create a TinyDuino Stack for your particular project without soldering
- The TinyDuino Processor Board is basically the Arduino Uno board shrunk down to the size of a quarter
- It uses the same Atmel Atmega328P microcontroller as used on the Uno
- The one functional difference between the TinyDuino Processor Board and the Arduino Uno is that the TinyDuino uses an 8MHz ceramic resonator instead of the 16MHz that the Uno uses.
 - There is a great reason for this - this allows the TinyDuino processor board to run down to 2.7 Volts, which enables it to run directly off of batteries (such as Li-ion and coin cells)

Normal Arduino - Pins are how we communicate with the world

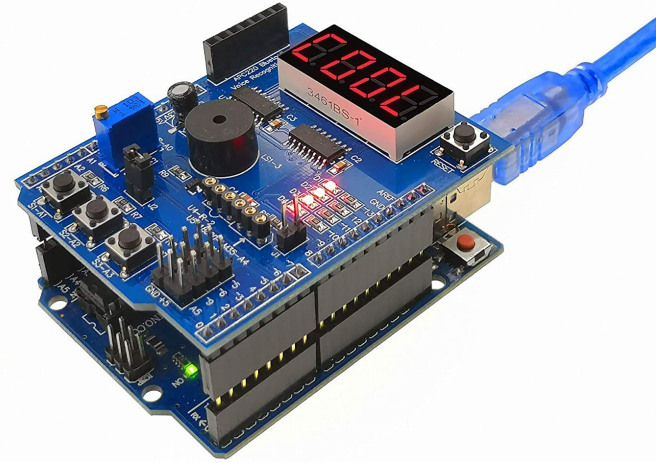
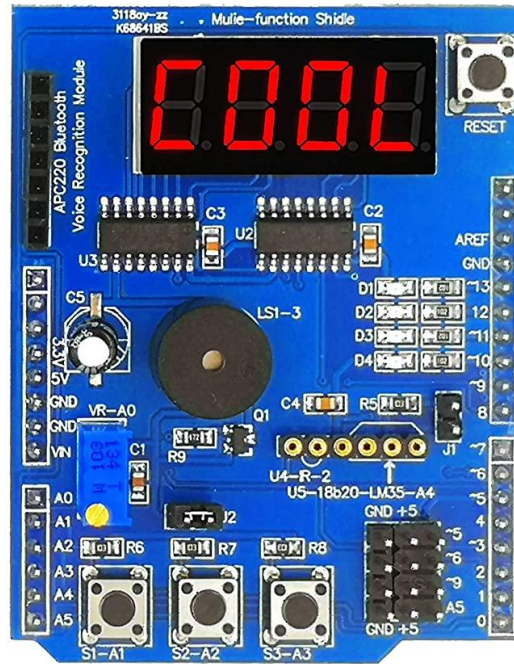


Input/output pins

Can Connect A “Shield” To Pins



Shield



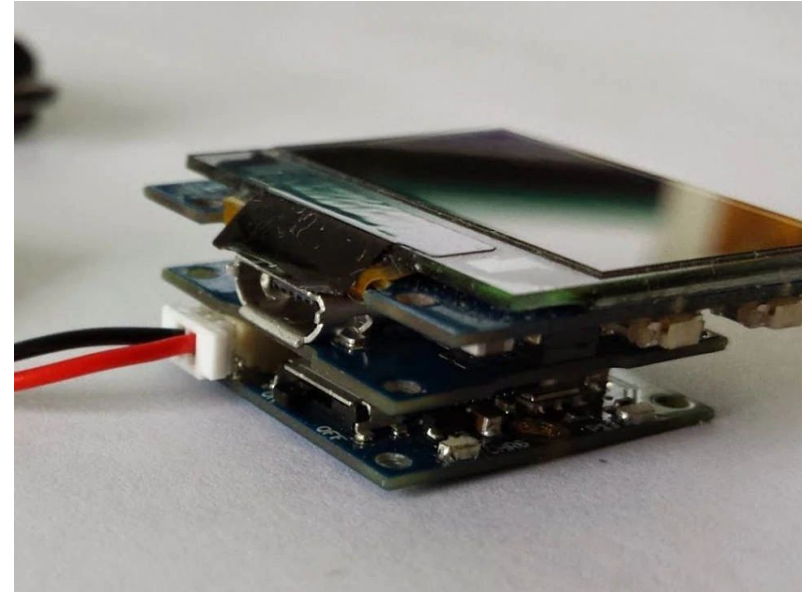
EEEEEE UNO R3 &
Multi-Function Shield Starter
kit on Amazon [here](#)

Tiny Watch - We Use Wirelings & Expansion Shields

- Expansion shields can stack through use of 32 pin port, similar to Arduino Shields



TinyCircuits. URL: <https://tinycircuits.com/products/tinyscreenplus>



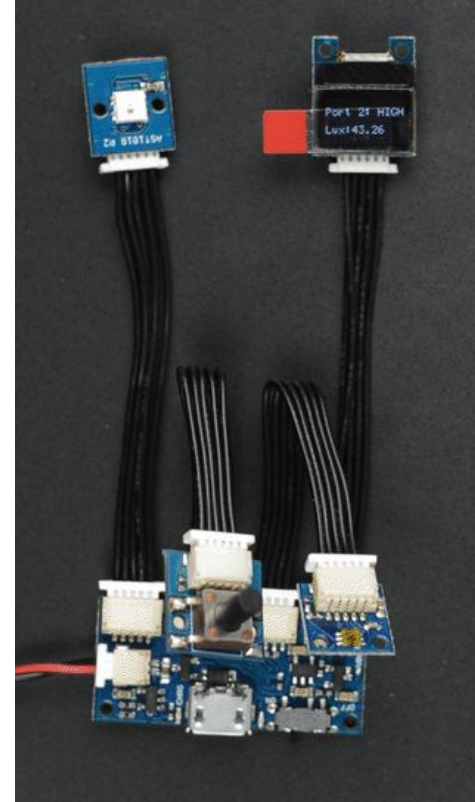
TinyCircuits. URL: https://tinycircuits.com/blogs/news/115742983-tinyscreen-assembly?_pos=7&_sid=88f65f6ef&_ss=r

Tiny Watch - We Use Wirelings & Expansion Shields

Text below edited from TinyCircuits - URL:

<https://tinycircuits.com/blogs/news/wireling-kickstarter-overview>

- Wirelings are tiny boards that plug into your projects to add sensors, I/Os, and tons of other possibilities!
- Without soldering, you can add Wirelings to your TinyCircuits projects with our Wiring Adapter TinyShield, or try out one of our new, specially designed processor boards
- You can also use Wirelings with the Standard Arduino using our Wiring Arduino Shield, or with a Raspberry Pi, using our Wiring Pi Hat
- All the magic happens in the 5-pin Wiring Cable that connects these versatile boards to your existing hardware platforms
- With the use of this cable, you can throw out that breadboard, and start prototyping right away by simply plugging in the components you want to use

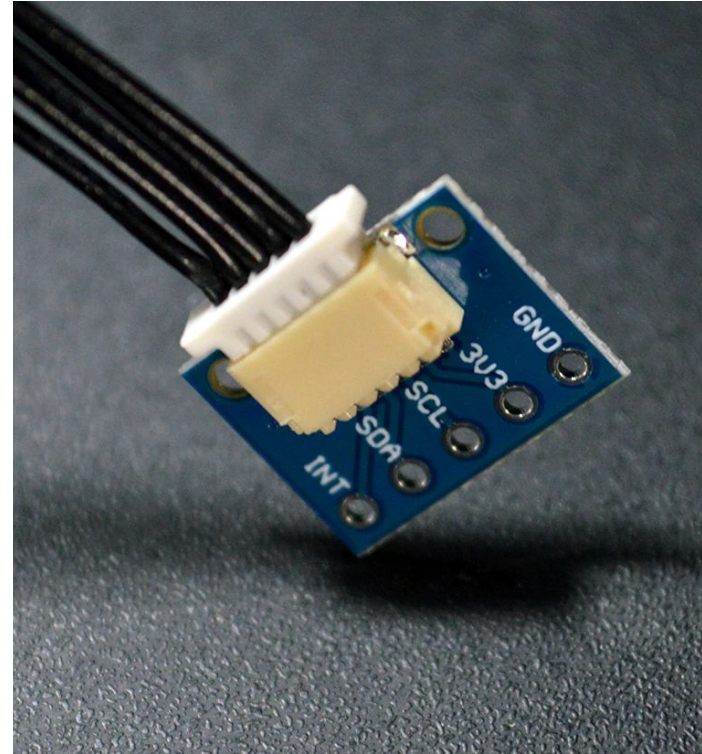


TinyCircuits. URL:

<https://tinycircuits.com/blogs/news/wireling-kickstarter-overview>

What's The Secret Behind Wirelings?

- Wiring cable contains 5 wires:
 - Power - 3V
 - Ground - 0V
 - Analog input/output
 - I2C - SDA
 - I2C - SCL
- I2C - Let's talk about this more



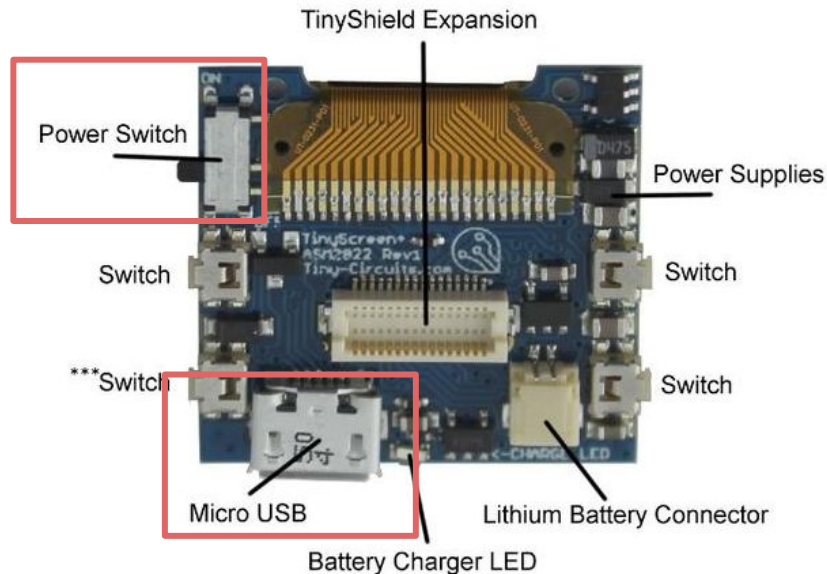
From TinyCircuits, cropped. URL:
<https://tinycircuits.com/pages/wiring-overview>

Tiny Watch & Pulse Oximeter

- Objectives:
 - Operate TinyScreen+
 - Upload example code (that comes with the library)
 - Observe variables, setup() and loop() and try correlating it with the output you see.
 - Pulse ox
 - Downloading library and putting it in the right folder
 - How to read a library (figuring out functions)

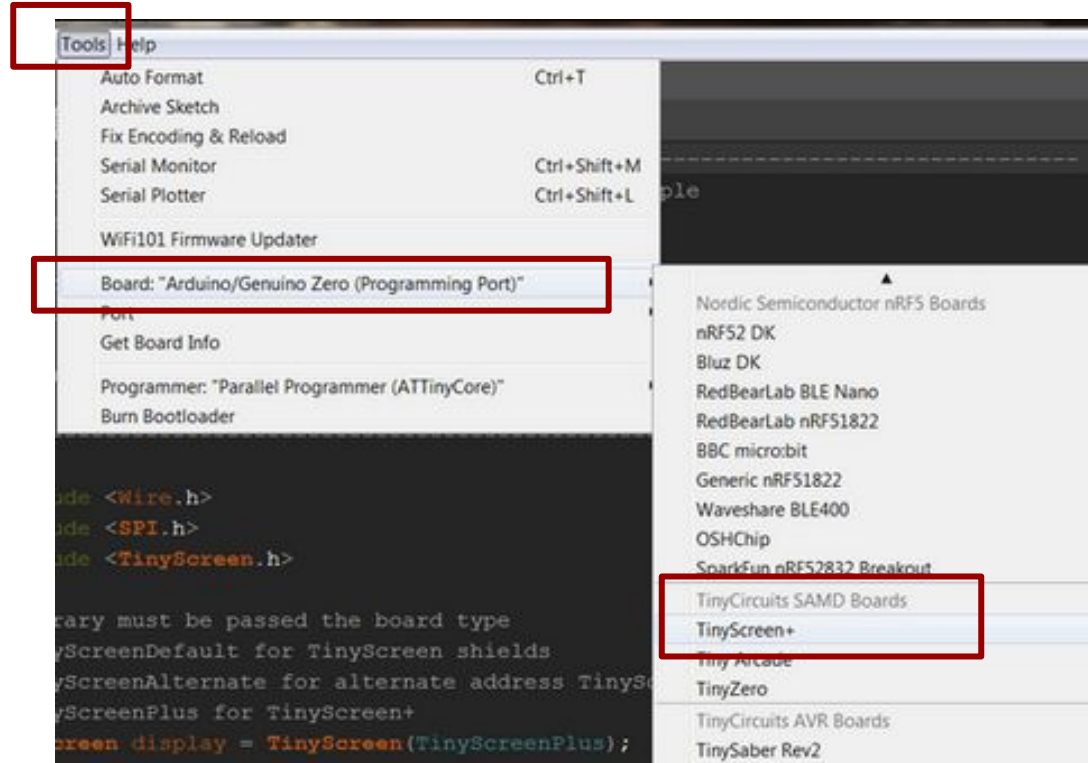
Connecting Arduino to TinyScreen

- Connect your **TinyScreen+** to the computer using the USB cable
- Make sure the **power switch** is toggled to “ON”



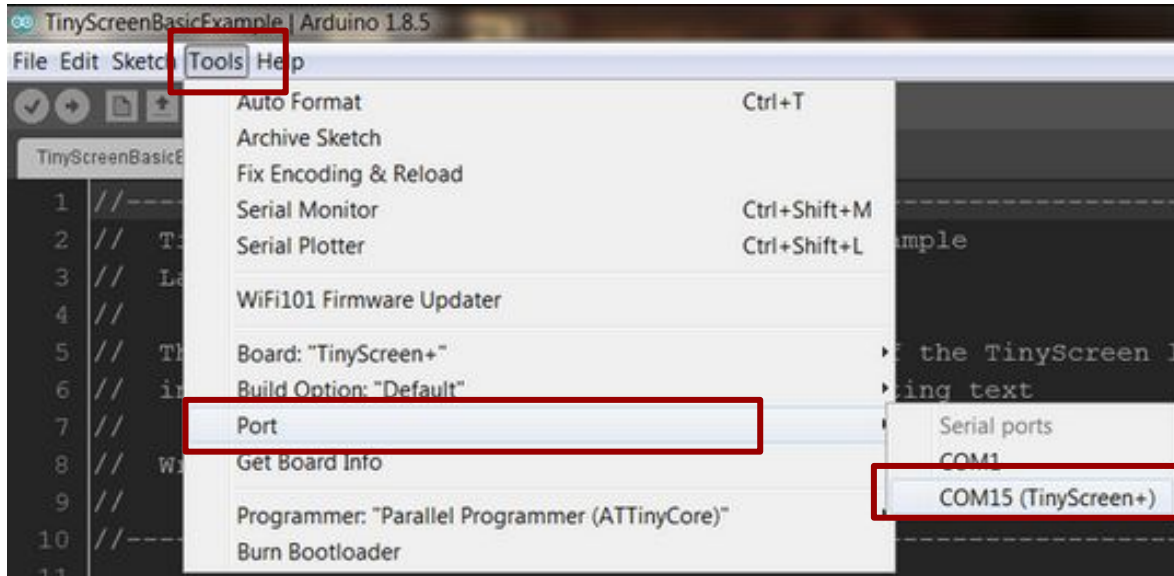
Connecting Arduino to TinyScreen

- Go to **Tools > Board** and find **TinyScreen+**



Connecting Arduino to TinyScreen

- Go to **Tools > Port** and find **COMx (TinyScreen+)**



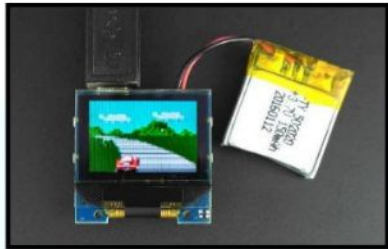
TinyScreen

- Open **File > Examples > TinyScreen > TinyScreenBasicExample**
- Upload the code to TinyScreen and see what it does!
- Remembering what we learned earlier, take a look at the **variables**, **setup()** and **loop()**

: will become particularly useful!

TinyScreen

- OLED
- What functions can you use?
 - Refresher on what a function is



Initialization and Control:

- TinyScreen()
- [startData\(\)](#)
- [endTransfer\(\)](#)
- begin()
- [on\(\)](#)
- [off\(\)](#)
- [setFlip\(\)](#)
- [setMirror\(\)](#)
- [setBitDepth\(\)](#)
- [setColorMode\(\)](#)
- setBrightness()

Accelerated Drawing Commands:

- [drawPixel\(\)](#)
- [drawLine\(\)](#)
- [drawRect\(\)](#)
- clearWindow()
- clearScreen()

Basic Graphics Commands:

- [writePixel\(\)](#)
- [writeBuffer\(\)](#)

- [setX\(\)](#)
- [setY\(\)](#)
- [goTo\(\)](#)

I²C and GPIO Related:

- getButtons()

Font:

- [setFont\(\)](#)
- [getFontHeight\(\)](#)
- [getPrintWidth\(\)](#)
- [setCursor\(\)](#)
- [fontColor\(\)](#)
- [write\(\)](#)
- [xMax = 95](#)
- [yMax = 63](#)

Internally Referenced Commands:

- [startCommand\(\)](#)
- [writeRemap\(\)](#)
- [writeGPIO\(\)](#)
- [print\(\)](#)

Pdf for detailed manual and function syntax:

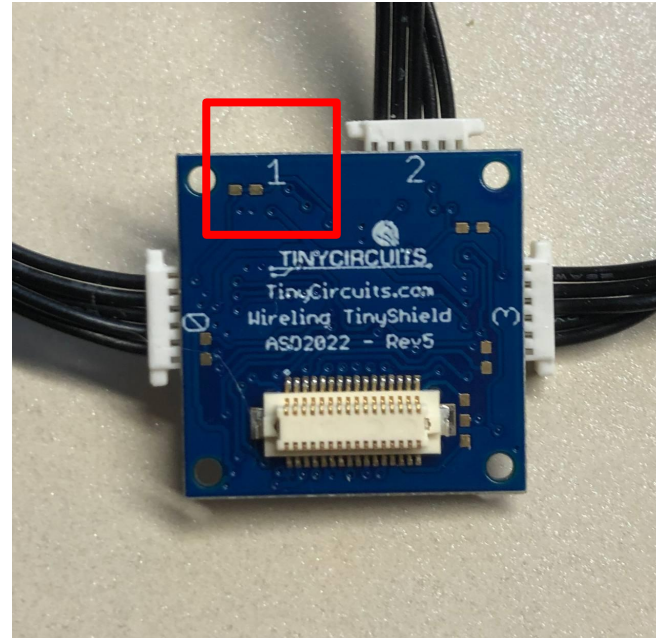
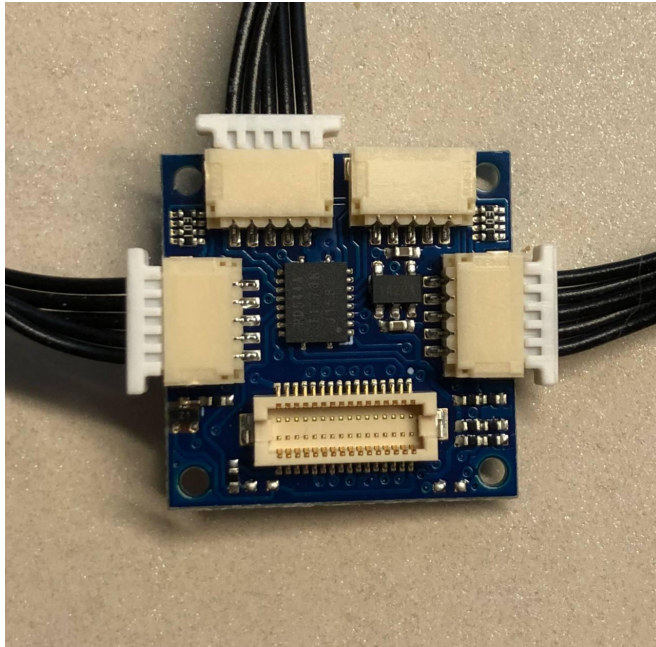
<https://usermanual.wiki/Pdf/TinyScreenReferenceManual.1153252474/view>

Let's sense our oxygen level using the pulse oximeter!

- Your pulse oximeter has to be aligned with a blood vessel if possible!
 - Readings won't be accurate otherwise (at least for the kit you got)

Let's sense our oxygen level using the pulse oximeter!

- Connect pulse oximeter to **port 1** of the wiring board

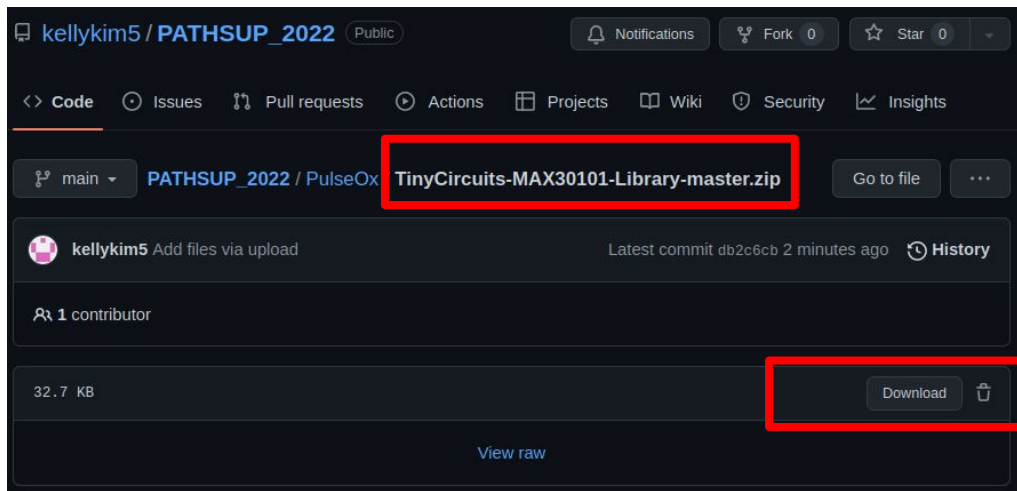


Let's sense our oxygen level using the pulse oximeter!

- Download library :

github.com/kellykim5/PATHSUP_2022/tree/main/PulseOx

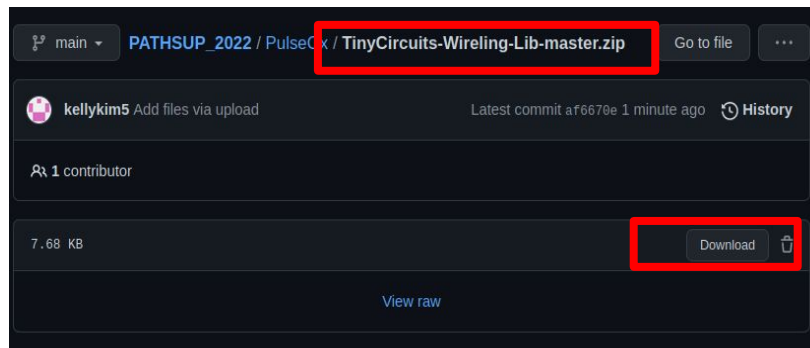
- Click
“TinyCircuits-MAX30101-Library-master.zip”
- Click “Download”
- Unzip



- Download library :

https://github.com/kellykim5/PATHSUP_2022/tree/main/PulseOx

- Click
“TinyCircuits-Wireling-Lib-master.zip”
- Click “Download”
- Unzip

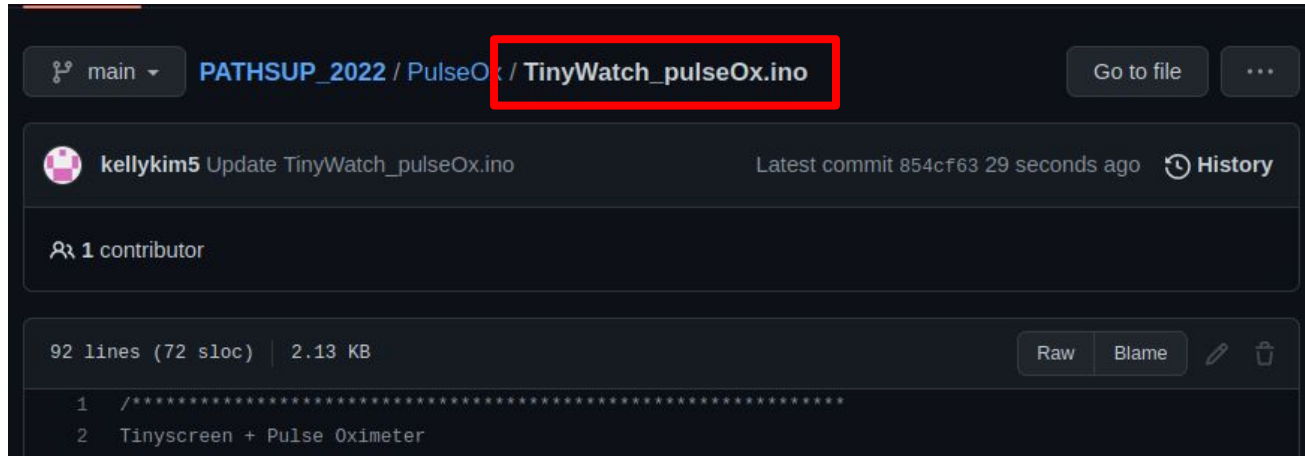


Let's sense our oxygen level using the pulse oximeter!

- Copy and paste the library folders that you just downloaded to
 - **Documents > Arduino > libraries**
 - You can just copy and paste whole folders

Let's sense our oxygen level using the pulse oximeter!

- Code
 - Click “TinyWatch_pulseOx.ino”
 - Copy the code and paste it in the Arduino IDE (**File > New**)



The screenshot shows a GitHub repository interface. At the top, the breadcrumb navigation path is 'PATHSUP_2022 / PulseOx / TinyWatch_pulseOx.ino', with the final file name highlighted by a red rectangular box. To the right of the path are buttons for 'Go to file' and a menu icon. Below the path, a commit message 'kellykim5 Update TinyWatch_pulseOx.ino' is displayed, along with the commit hash '854cf63' and the time '29 seconds ago'. A 'History' link is also present. Underneath, it shows '1 contributor'. At the bottom, the file statistics '92 lines (72 sloc) | 2.13 KB' are shown, followed by 'Raw' and 'Blame' buttons and icons for editing and deleting the file. The code editor area shows the first two lines of the file: '1 /*****' and '2 Tinyscreen + Pulse Oximeter'.

Let's sense our oxygen level using the pulse oximeter!

- Save the code to
 - **Documents > Arduino**
- Name of the file can be whatever you want!
 - For example, you can save it as **Tinywatch_pulseOx**
- Connect your tinscreen to the computer using the USB cable
- Check board and port on “Tools” tab
- Upload the code and see what it does!